Valid Parentheses with a Stack

Topic: CompetitiveProgramming File: c6.pdf

PROBLEM

Determine whether a string containing (), [], and {} is correctly balanced and nested.

APPROACH

Push each opening bracket. For a closing bracket, the stack must be nonempty and its top must be the matching opening bracket; otherwise reject. At the end, the stack must be empty.

EXAMPLE

{[()]} is valid. ([)] is invalid because the closing brackets do not match the most recent opening bracket.

COMPLEXITY

Time: O(n), as each character is processed once. Extra space: O(n) in the worst case.

EDGE CASES

The empty string is balanced under the usual definition. A closing bracket with an empty stack is immediately invalid.